AZ-Eval: A Parallel Azerbaijani–English Benchmark Reveals Negative Cross-Lingual Transfer from Kazakh Fine-Tuning

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CMAI 2026 · Nazarbayev University, Astana

Open multilingual models are measured on high-resource languages

Kazakh has open evaluation sets - freshqa_kazakh, defan_kazakh, BeyneleBench

Azerbaijani, to our knowledge, has none

Both are Turkic. Kazakh is written in Cyrillic, Azerbaijani in Latin.

The question

Does fine-tuning on Kazakh transfer to Azerbaijani?

Kazakh is the closest well-resourced Turkic relative of Azerbaijani.

Adapting to a related language is the intuitive move for a low-resource language.

AZ-Eval - an open parallel benchmark

verified_by is a gate, not metadata: unverified rows never enter the set.

356 parallel AZ/EN short-answer items	255 hand-written
Every item human-verified	101 from templat
7 categories	math, history, geography, science, cultu
Accepted answers expanded	rule-based Azer

Four nested normalizations

STRICT → MORPH → LENIENT → TRANSLIT

suffix stripping · diacritic folding · Cyrillic→Latin

Each stage adds exactly one transformation, applied symmetrically to prediction and reference - so each score increment attributes error mass
to that single surface phenomenon.

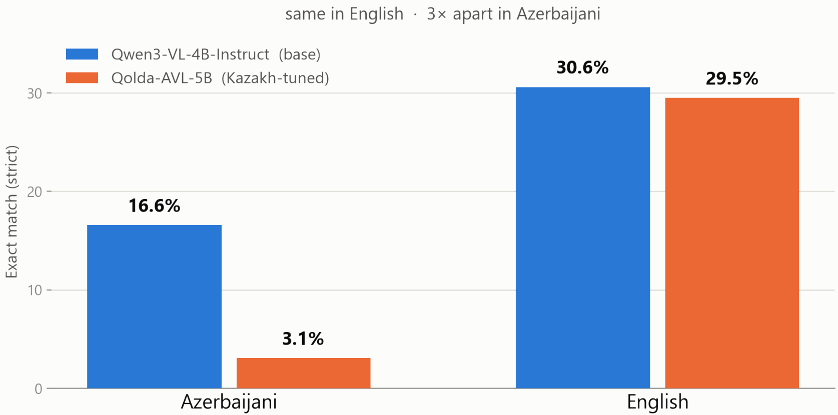
Paired comparisons · 1000-resample bootstrap · permutation test · Holm correction

Result 1 - the gap is universal

Majority-class baseline: 1.1%

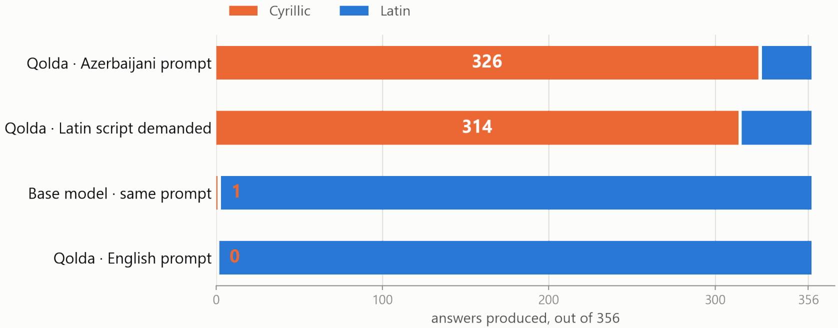
Model	AZ	EN	Gap	p (Holm)
Qwen3-1.7B	5.9%	23.6%	17.7	0.0016
Qwen3-VL-4B-Instruct	16.6%	30.6%	14.0	0.0016
issai/Qolda-AVL-5B	3.1%	29.5%	26.4	0.0016

Result 2: transfer is NEGATIVE



Holm-corrected p = 0.0024 · same architecture, same quantization, same prompt

The mechanism is orthographic



Understands the question, finds the answer, writes it in the wrong alphabet.

Is it simply a weaker model?

Indistinguishable in English. 35.4 points apart in Azerbaijani.

Language	Base model	Kazakh-tuned	Gap	p (Holm)
English	59.4%	59.4%	0.0	1.0000
Azerbaijani (STRICT)	45.8%	10.4%	35.4	0.0030

96 world + science items - facts every model demonstrably knows in English

Takeaway

Dataset CC BY 4.0 · Code MIT ·
github.com/nihatgaribli/AZ-Eval

1. Fine-tuning on a closely related language can HARM rather than help

- when the writing systems differ.

2. Most of the damage is orthographic, not semantic

- transliteration removes 56% of the deficit.

3. Which means it is in principle fixable

- e.g. by constraining the output script at decoding time.

Backup slides

Backup - where the gap is widest

Mathematics: knowledge is present in English, the language blocks it.

Category	n	Qwen3-VL-4B (AZ / EN)	Qolda-AVL-5B (AZ / EN)
mathematics	28	3.6% / 25.0%	3.6% / 28.6%
world	45	51.1% / 57.8%	0.0% / 60.0%
history	45	4.4% / 0.0%	0.0% / 2.2%

Backup - what survives transliteration

Transliteration removes 56% of the deficit - but not all of it.

Residual after TRANSLIT: 5.9 points, Holm $p = 0.0364$ - still significant.

So the claim is: the damage is LARGELY orthographic, not purely orthographic.

Backup - how the dataset was built

255 items hand-written by a native Azerbaijani speaker

101 from 8 templated SPARQL queries × 4 syntactic variants (round-robin)

No LLM authored, translated, or answered any dataset item.

Rejected candidates are retained, so the acceptance rate stays auditable.

Items whose gold answer depends on a contested position are excluded (6 removed).

Backup - limitations

n = 356. Confidence intervals are ±2–5 points; every reported comparison
survives Holm correction.

One model pair. The 9B/8B pair did not fit in 8 GB of VRAM.

Two knowledge regimes are mixed on purpose - and kept in separate categories
so the distinction stays visible instead of averaging away.

Backup - answer extraction

Identical rules for both languages:

first line only · strip ’Cavab:’ / ’Answer:’ prefixes

strip markdown emphasis and trailing punctuation · remove thinking blocks

Raw model outputs are kept untouched (8 runs), so the scoring rule can be
changed and the analysis re-run without spending another GPU-hour.